



Caringbah High School

2015

Trial HSC Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen (Black pen is preferred)
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2 – 5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6 – 14

90 marks

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Question 1 - 10 (1 mark each) Answer on page provided.

1 The argument of $z = -3i$ is:

- A) $\frac{\pi}{2}$ B) $\frac{3\pi}{2}$ C) $-\frac{\pi}{2}$ D) $-\frac{3\pi}{2}$

2 Consider a polynomial $P(x)$ that has real coefficients. Which of the following could not be a pair of possible solutions to $P(x) = 0$?

- A) $x_1 = -1 + i; x_2 = -1 - i$ B) $x_1 = 1 + i; x_2 = 1 - i$
- C) $x_1 = -2 + i; x_2 = 2 - i$ D) $x_1 = -2 + i; x_2 = -2 - i$

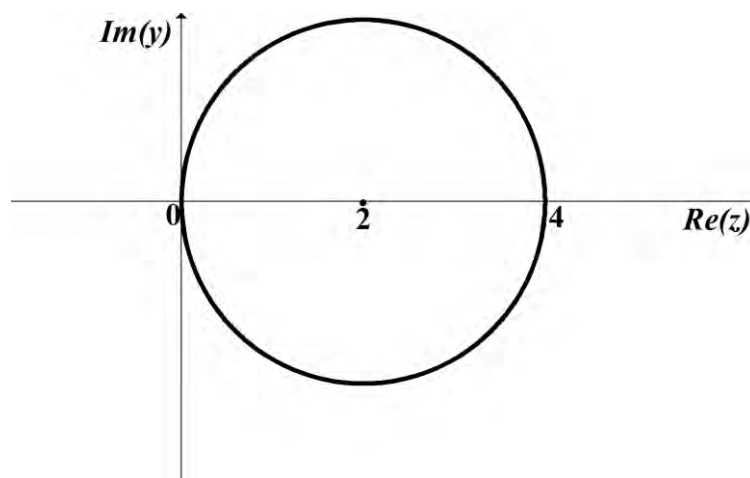
3 Given that $z^3 = 1$ and ω is a complex solution, the value of $\omega^3 + \omega^4 + \omega^5$ is:

- A) 0 B) 1
- C) ω D) $-\omega$

4 The coordinates of the foci of the hyperbola $\frac{x^2}{4} - y^2 = 1$ are given by:

- A) $(0, \pm\sqrt{5})$ B) $(\pm\sqrt{5}, 0)$
- C) $(0, \pm2\sqrt{5})$ D) $(\pm2\sqrt{5}, 0)$

- 5 Which of the following is the equation of the circle shown below?



- A) $(z + 2)(\bar{z} + 2) = 4$ B) $(z - 2)(\bar{z} - 2) = 4$
- C) $(z + 2i)(\bar{z} - 2i) = 4$ D) $(z - 2i)(\bar{z} + 2i) = 4$

- 6 The roots of the polynomial $4x^3 + 4x - 5 = 0$ are α, β and γ .

What is the value of $(\beta + \gamma - 3\alpha)(\alpha + \beta - 3\gamma)(\gamma + \alpha - 3\beta)$?

- A) 16 B) 80
- C) -16 D) -80
- 7 The equation $x^2 + 2y^2 - 2xy + x = 8$ defines y implicitly as a function of x .

What is the value of $\frac{dy}{dx}$ at the point $(3, 2)$?

- A) $\frac{1}{4}$ B) $-\frac{1}{4}$
- C) $\frac{3}{2}$ D) $-\frac{3}{2}$

- 8 All of the integrals below are of the form $\int_{-1}^1 f(x) dx$.

Without evaluating, which of the integrals can be rewritten as $2\int_0^1 f(x) dx$?

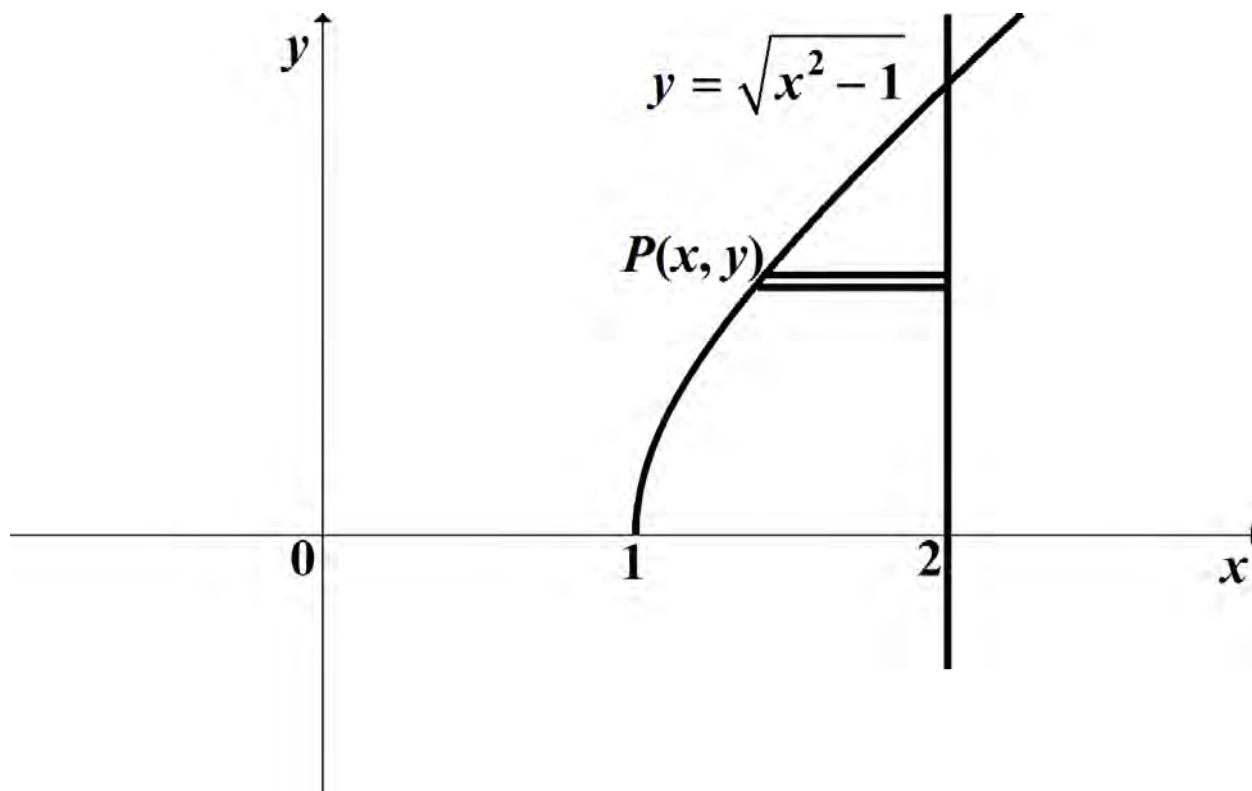
- A) $\int_{-1}^1 e^x \tan^{-1}(x^2) dx$ B) $\int_{-1}^1 \frac{x^2 \sin x}{x^2 + 5} dx$
- C) $\int_{-1}^1 \sqrt{x^2 + e^x} dx$ D) $\int_{-1}^1 x^3 \sin^{-1} x dx$

- 9 Using the recurrence relation $U_n = \int \tan^n x dx = \frac{\tan^{n-1} x}{n-1} - U_{n-2}$

what is the value of $\int \tan^6 x dx$?

- A) $\frac{\tan^5 x}{5} - \frac{\tan^4 x}{4} + \frac{\tan^3 x}{3} - \frac{\tan^2 x}{2} + \tan x + c$
- B) $\frac{\tan^5 x}{5} - \frac{\tan^3 x}{3} + \tan x + c$
- C) $\frac{\tan^5 x}{5} - \frac{\tan^3 x}{3} + \tan x - x + c$
- D) $\frac{\tan^5 x}{5} - \frac{\tan^3 x}{3} - \tan x - x + c$

10



The region bounded by the x -axis, the curve $y = \sqrt{x^2 - 1}$ and the line $x = 2$ is rotated around the y -axis.

The slice at $P(x, y)$ on the curve is perpendicular to the axis of rotation.

What is the volume δV of the annular slice formed?

- | | |
|---------------------------|--|
| A) $\pi(3 - y^2)\delta y$ | B) $\pi\left(4 - (y^2 + 1)^2\right)\delta y$ |
| C) $\pi(4 - x^2)\delta x$ | D) $\pi(2 - x^2)\delta x$ |

END OF MULTIPLE CHOICE QUESTIONS

Section II**60 marks****Attempt all questions 11–14****Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.**Marks**a) Let $z = 1 + i$ and $u = 2 - i$. Find:

(i) $\operatorname{Im}(uz)$. 1

(ii) $|u - z|$. 1

(iii) $-i\bar{u}$. 1

b) Evaluate $\int_0^1 x^2 \sqrt{1-x} \, dx$. 3c) It is given that $Z = \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}i$.

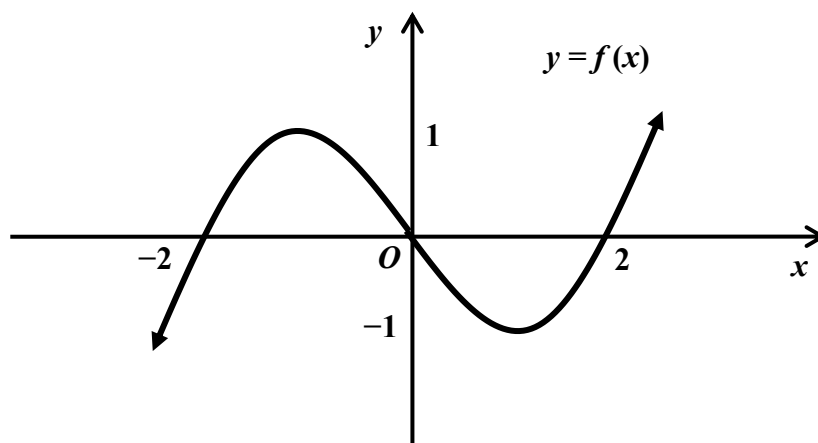
(i) Show that Z can be expressed in the form $\sqrt{2} \operatorname{cis} \frac{\pi}{6}$. 2

(ii) Hence, express Z^8 in $x + iy$ form. 2

d) (i) Prove that if the polynomial $P(x)$ has a root α of multiplicity m 2then $P'(x)$ has a root α of multiplicity $m - 1$.(ii) Hence, given that the polynomial $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$ has 3a root of multiplicity 3, find all the roots of $P(x)$.

Question 12 (15 marks) Start a NEW booklet.**Marks**

- a) Consider the function $y = f(x)$ drawn below.



On separate sketches, showing any important features, neatly draw the graphs of

- | | | |
|-------|--|---|
| (i) | $y = 1 - f(x)$ | 1 |
| (ii) | $y = \sqrt{f(x)}$ | 1 |
| (iii) | $y = f\left(\frac{1}{x}\right)$ | 2 |
| (iv) | $y = e^{f(x)}$ | 2 |
| | | |
| b) | Evaluate $\int_8^{12} \frac{1}{x^2 - 16x + 80} dx$ | 3 |

Question 12 continues on page 8

Question 12 (continued)

Marks

c) The equation $x^3 + 2x - 1 = 0$ has roots α , β and γ . Find

(i) the value of $\alpha^2 + \beta^2 + \gamma^2$. 2

(ii) the value of $\alpha^3 + \beta^3 + \gamma^3$. 2

(iii) the equation with roots $-\alpha, -\beta, -\gamma$. 2

End of Question 12

Question 13 (15 marks) Start a NEW booklet.**Marks**

- a) Consider the locus of the point $P(x, y)$ whose co-ordinates satisfy the parametric equations $x = \cos t$, $y = 1 - \cos 2t$.

- (i) State the domain and range of the locus of P . 2
- (ii) Determine the Cartesian equation of the locus of P . 2
- (iii) Neatly sketch the curve traced by the point $P(x, y)$. 1

- b) Find $\int \frac{\cos x}{\sin x + \sin^2 x} dx$. 4

- c) An ellipse has the equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$

- (i) Prove that the tangent to the ellipse at $P(4\cos\theta, 3\sin\theta)$ has 3

equation $\frac{x \cos \theta}{4} + \frac{y \sin \theta}{3} = 1$.

- (ii) The ellipse meets the y -axis at B and B' . The tangents at B and B' meet the tangent at P at the points Q and Q' . 3

Find $BQ \times B'Q'$.

Question 14 (15 marks) Start a NEW booklet.**Marks**

a) Consider the functions $f(x) = |x| + 1$ and $g(x) = \frac{6}{|x|}$.

(i) Solve the equation $f(x) = g(x)$. 2

(ii) Sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same axes. 2

(iii) Hence or otherwise solve the inequality $g(x) > f(x)$. 1

b) i) Prove that $a^2 + b^2 \geq 2ab$. 1

ii) Hence or otherwise prove that $(p + 2)(q + 2)(p + q) \geq 16pq$ 2
where p and q are positive real numbers.

c) The base of a solid is the region between the lines $y = 3x$ and $y = -x$ from $x = 0$ to $x = 2$.

Each cross section by planes perpendicular to the x -axis is a square.

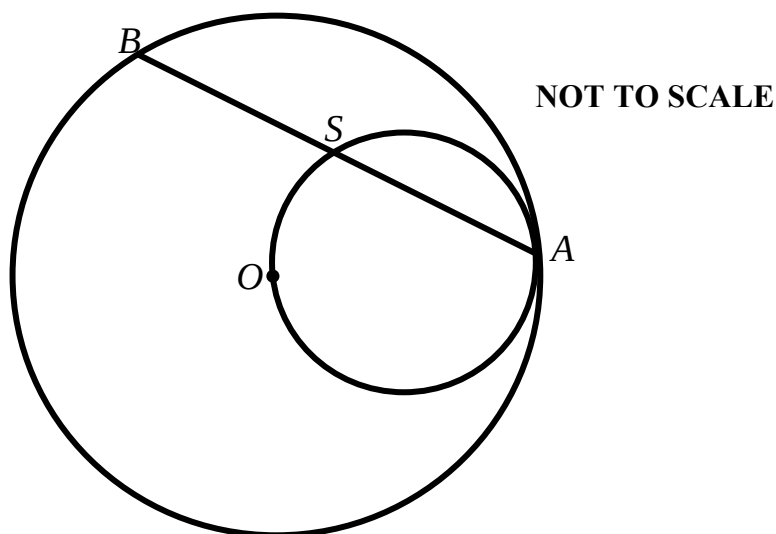
Calculate the volume of the solid. 3

Question 14 continues on page 11

Question 14 (continued)

Marks

- d) In the diagram below the two circles touch internally at A . O is the centre of the larger circle. B is a point on the larger circle and chord AB cuts the smaller circle at S .



Answer this question on the page provided
--

Prove that chord AB is bisected at S .

4

[Ensure that any constructions are clearly shown and labelled]

End of Question 14

Question 15 (15 marks) Start a NEW booklet.**Marks**

- a) Let z be a complex number such that $|z| = r$ and $\arg z = \theta$ for $0 < \theta < \frac{\pi}{2}$.

Prove that $\arg(r^2 - z^2) = \theta - \frac{\pi}{2}$.

3

- b) (i) Show that $\frac{d}{dx} \left(\frac{x^2}{2} \ln x - \frac{x^2}{4} \right) = x \ln x$

1

- (ii) Using $\frac{1}{3}$ of a page neatly sketch the region in the number plane that contains all points satisfying simultaneously the inequalities

$$x \leq 1, y \geq 1, y \leq e^x.$$

1

- (iii) This region is rotated through one complete revolution about the x -axis.

Use the method of cylindrical shells to find the volume of the resulting solid.

4

- c) If n is a positive integer and $f(x) = e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right)$, $x \geq 0$

- (i) Show that $f(x)$ is a decreasing function.

3

- (ii) Deduce that for $x > 0$ and n any positive integer,

3

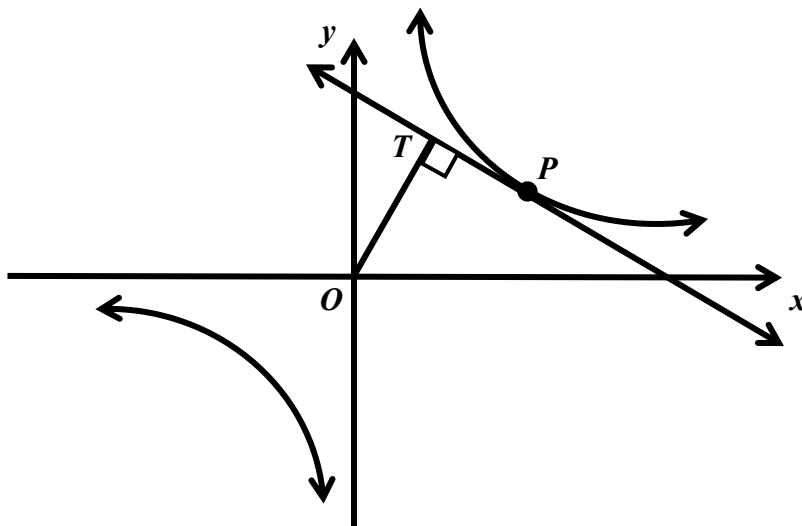
$$e^x \geq 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$$

Question 16 (15 marks) Start a NEW booklet.**Marks**

a) Find $\int \frac{e^{2x}}{e^x - 1} dx$.

3

b)



The point $P\left(cp, \frac{c}{p}\right)$ lies on the hyperbola $xy = c^2$. The point T lies at the foot of the perpendicular drawn from the origin O to the tangent at P .

(i) Show that the tangent at P has equation $x + p^2y = 2cp$.

2

(ii) If the coordinates of T are (x_1, y_1) show that $y_1 = p^2x_1$.

1

(iii) Show that the locus of T is given by $(x^2 + y^2)^2 = 4c^2xy$.

2

Question 16 continues on page 14

Question 16 (continued)

Marks

c) Consider the integral $I_n = \int_0^1 x^{2n+1} e^{-x^2} dx$.

(i) Use integration by parts to show that $I_n = -\frac{1}{2e} + nI_{n-1}$, for $n \geq 1$. 2

(ii) Show that $I_0 = \frac{1}{2} - \frac{1}{2e}$ and $I_1 = \frac{1}{2} - \frac{1}{e}$ 1

(iii) Prove by mathematical induction that for all $n \geq 1$: 3

$$1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} = e - \frac{2eI_n}{n!}$$

(iv) It is given that $0 \leq I_n \leq 1$ because $0 \leq x^{2n+1} e^{-x^2} \leq 1$, for $0 \leq x \leq 1$.

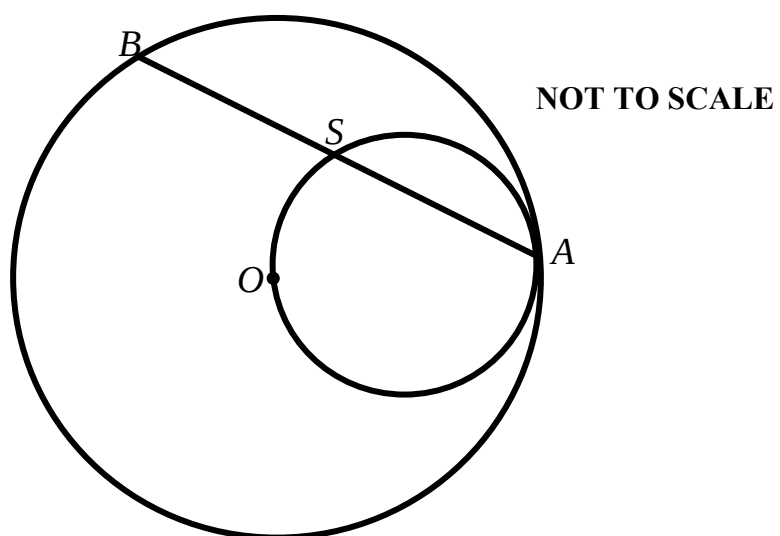
[Do not prove this]

Use this fact to help to evaluate $\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$ 1

giving your answer in exact form.

END OF EXAM

Question 14d (additional diagram) Candidate Name/Number: _____



Hand in this page inside the Question 14 booklet

[illegible]

Multiple Choice Section:

1.C 2.C 3.A 4.B 5.B
 6.D 7.D 8.D 9.C 10.A

Question 1.

$$-\frac{\pi}{2} \quad \text{-----} \quad \boxed{C}$$

Question 2.

Solutions need to be complex conjugates as coefficients are real

$$x_1 = -2 + i; x_2 = 2 - i \text{ are not } \text{-----} \quad \boxed{C}$$

Question 3.

$$\omega^3 - 1 = 0 \rightarrow (\omega - 1)(1 + \omega + \omega^2) = 0$$

and since $\omega \neq 1$ (must be non-real), then

$$\omega^3 + \omega^4 + \omega^5 = 1 + \omega + \omega^2$$

$$= 0 \quad \text{-----} \quad \boxed{A}$$

Question 4.

$$a=2; b=1; b^2 = a^2(e^2 - 1); \text{Foci}(\pm ae, 0)$$

$$\therefore e^2 = \frac{5}{4} \rightarrow e = \frac{\sqrt{5}}{2}$$

$$\therefore \text{Foci}(\pm\sqrt{5}, 0) \quad \text{-----} \quad \boxed{B}$$

Question 5.

$$\text{Equation in Cartesian form is } (x-2)^2 + y^2 = 4$$

$$(z-2)(\bar{z}-2) = 4 \rightarrow z\bar{z} - 2(z+\bar{z}) + 4 = 4$$

$$\therefore x^2 + y^2 - 2(2x) + 4 = 4$$

$$\therefore x^2 - 4x + 4 + y^2 = 4$$

$$\therefore (x-2)^2 + y^2 = 4 \quad \text{-----} \quad \boxed{B}$$

Question 6.

$$(\beta + \gamma - 3\alpha)(\alpha + \beta - 3\gamma)(\gamma + \alpha - 3\beta)$$

$$= (\alpha + \beta + \gamma - 4\alpha)(\alpha + \beta + \gamma - 4\beta)(\alpha + \beta + \gamma - 4\gamma)$$

$$\text{Now } \alpha + \beta + \gamma = -\frac{b}{a} = 0 \text{ and } \alpha\beta\gamma = \frac{c}{a} = \frac{5}{4}$$

$$\therefore = (-4\alpha)(-4\beta)(-4\gamma)$$

$$= -64\alpha\beta\gamma$$

$$= -64 \times \frac{5}{4} = -80 \quad \text{-----} \quad \boxed{D}$$

Question 7.

$$x^2 + 2y^2 - 2xy + x = 8$$

$$\therefore 2x + 4y \frac{dy}{dx} - 2\left(x \times \frac{dy}{dx} + y \times 1\right) + 1 = 0$$

$$\therefore 2(2y - x) \frac{dy}{dx} = 2y - 2x - 1$$

$$\therefore \frac{dy}{dx} = \frac{2y - 2x - 1}{2(2y - x)} \text{ and at } P(3, 2)$$

$$\frac{dy}{dx} = \frac{4 - 6 - 1}{2(4 - 3)} = -\frac{3}{2} \quad \text{-----} \quad \boxed{D}$$

Question 8.

$$\text{Let } f(x) = x^3 \sin^{-1} x$$

$$\begin{aligned} \therefore f(-x) &= (-x)^3 \sin^{-1}(-x) \\ &= -x^3 \times -\sin^{-1} x \\ &= x^3 \sin^{-1} x = f(x) \end{aligned}$$

hence $f(x)$ is an even function

hence ----- $\boxed{D}$

Question 9.

$$U_6 = \frac{\tan^5 x}{5} - U_4$$

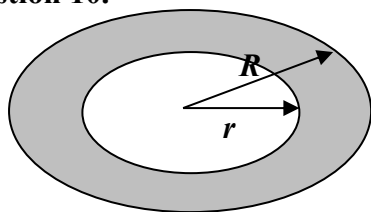
$$U_4 = \frac{\tan^3 x}{3} - U_2$$

$$U_2 = \frac{\tan x}{1} - U_0$$

$$U_0 = \int 1 dx = x + c$$

$$\therefore U_6 = \frac{\tan^5 x}{5} - \frac{\tan^3 x}{3} + \tan x - x + c \quad \text{--- [C]}$$

Question 10.



where $R = 2$
and $r = x$

$$\begin{aligned} \therefore \text{Area of slice} &= \pi(2^2 - x^2) \\ &= \pi(4 - (y^2 + 1)) \end{aligned}$$

$$\therefore \text{Volume of slice} = \pi(3 - y^2) \delta y \quad \text{----- [A]}$$

Question 11

a) i) $\text{Im}[(1+i)(2-i)] = \text{Im}(3+1i) = 1$

ii) $|u - z| = |1 - 2i| = \sqrt{5}$

iii) $-i\bar{u} = -i \times (2+i) = 1 - 2i$

b) Let $u = 1 - x \rightarrow x = 1 - u$ and $du = -dx$
When $x = 0, u = 1$; $x = 1, u = 0$.

$$\therefore I = -\int_1^0 (1-u)^2 \sqrt{u} du$$

$$= \int_0^1 u^{\frac{1}{2}} - 2u^{\frac{3}{2}} + u^{\frac{5}{2}} du$$

$$= \left[\frac{2}{3} u^{\frac{3}{2}} - \frac{4}{5} u^{\frac{5}{2}} + \frac{2}{7} u^{\frac{7}{2}} \right]_0^1$$

$$= \frac{2}{3} - \frac{4}{5} + \frac{2}{7} = \frac{16}{105}$$

c) i) $|Z| = r = \sqrt{\frac{6}{4} + \frac{2}{4}} = \sqrt{2}$

$$\arg Z = \theta = \tan^{-1} \left(\frac{\sqrt{2}/2}{\sqrt{6}/2} \right)$$

$$\therefore \theta = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\pi}{6}$$

ii) $Z^8 = \left(\sqrt{2} \text{cis} \left(\frac{\pi}{6} \right) \right)^8$
 $= 2^4 \text{cis} \left(\frac{8\pi}{6} \right) \quad \text{DMT}$

$$= 16 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right)$$

$$= 16 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = -8(1 + \sqrt{3}i)$$

d) i) Let $P(x) = (x - \alpha)^m Q(x)$

$$\therefore P'(x) = m(x - \alpha)^{m-1} Q(x) + (x - \alpha)^m Q'(x)$$

$$= (x - \alpha)^{m-1} (mQ(x) + (x - \alpha)Q'(x))$$

hence $P'(x)$ has a root α of multiplicity $m - 1$.

ii) $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$

$$P'(x) = 4x^3 + 3x^2 - 6x - 5$$

$$P''(x) = 12x^2 + 6x - 6$$

$$P''(x) = 0 \rightarrow 2x^2 + x - 1 = 0$$

$$\therefore (2x - 1)(x + 1) = 0 \rightarrow x = \frac{1}{2} \text{ or } x = -1$$

when $x = -1, P'(-1) = -4 + 3 + 6 - 5 = 0$

when $x = -1, P(-1) = 1 - 1 - 3 + 5 - 2 = 0$

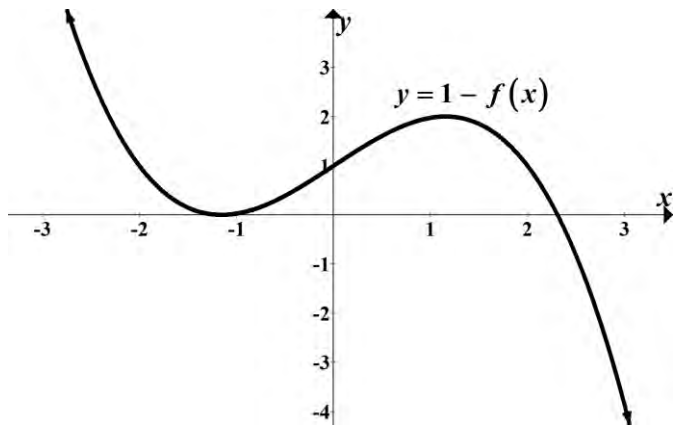
hence $x = -1$ is a root of multiplicity 3

$$\therefore P(x) = (x + 1)^3 (x - 2) \text{ by inspection}$$

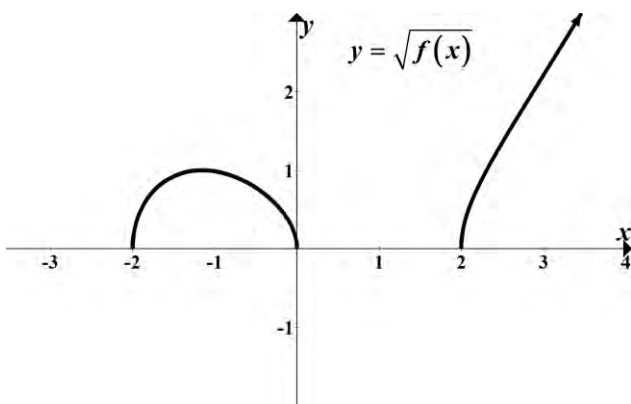
hence $P(x)$ has roots $x = -1, -1, -1, 2$.

Question 12.

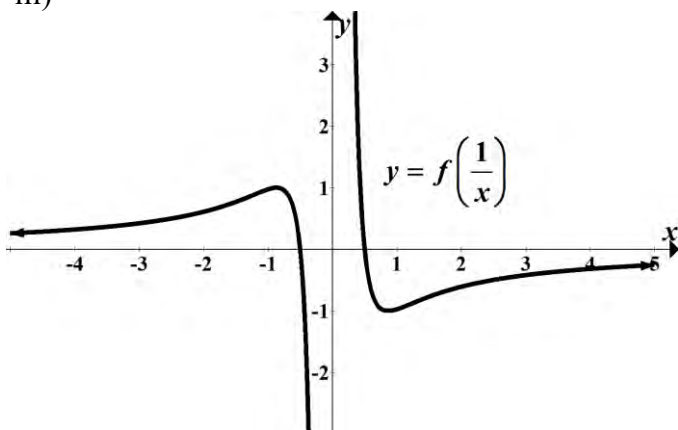
a) i)



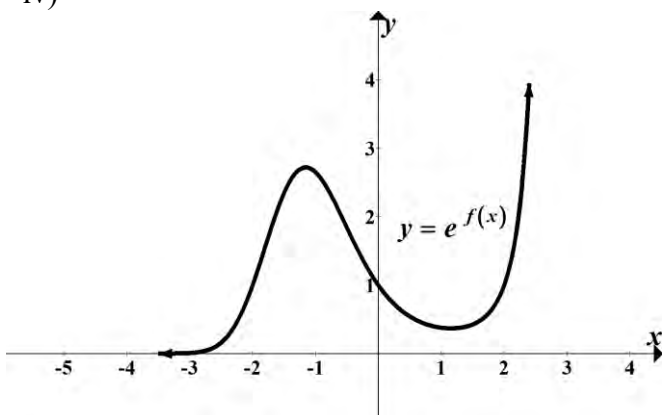
ii)



iii)



iv)



$$\begin{aligned} \text{b) } I &= \int_8^{12} \frac{1}{(x^2 - 16x + 64) + 16} dx \\ &= \int_8^{12} \frac{1}{(x-8)^2 + 4^2} dx \\ &= \frac{1}{4} \left[\tan^{-1} \left(\frac{x-8}{4} \right) \right]_8^{12} \\ &= \frac{1}{4} (\tan^{-1} 1 - \tan^{-1} 0) = \frac{\pi}{16} \end{aligned}$$

c) i)

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= (0)^2 - 2(2) = -4 \end{aligned}$$

$$\text{ii) } x^3 + 2x - 1 = 0 \rightarrow x^3 = 1 - 2x$$

$$\therefore \alpha^3 = 1 - 2\alpha$$

$$\beta^3 = 1 - 2\beta$$

$$\gamma^3 = 1 - 2\gamma$$

$$\begin{aligned} \therefore \alpha^3 + \beta^3 + \gamma^3 &= 3 - 2(\alpha + \beta + \gamma) \\ &= 3 - 2(0) = 3 \end{aligned}$$

$$\text{iii) Let } x = -\alpha \rightarrow \alpha = -x$$

$\therefore$ the required equation is given by

$$\therefore (-x)^3 + 2(-x) - 1 = 0$$

$$\therefore -x^3 - 2x - 1 = 0$$

$$\therefore x^3 + 2x + 1 = 0$$

Question 13.

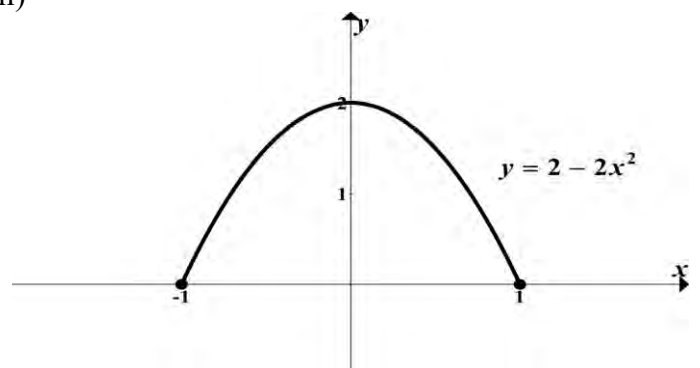
$$\text{a) i) D: } -1 \leq x \leq 1; \text{ R: } 0 \leq y \leq 2$$

$$\text{ii) } y = 1 - (2\cos^2 t - 1)$$

$$= 2 - 2(\cos t)^2$$

$$\therefore y = 2 - 2x^2$$

iii)



$$13b) I = \int \frac{\cos x}{\sin x + \sin^2 x} dx$$

$$\text{Let } u = \sin x \rightarrow du = \cos x dx$$

$$\therefore I = \int \frac{1}{u + u^2} du = \int \frac{1}{u(1+u)} du$$

$$\text{Now } \frac{1}{u(1+u)} = \frac{1}{u} - \frac{1}{u+1} \text{ using partial fractions}$$

$$\therefore I = \int \frac{1}{u} - \frac{1}{u+1} du$$

$$= \ln u - \ln(1+u)$$

$$= \ln\left(\frac{u}{1+u}\right)$$

$$\therefore I = \ln\left(\frac{\sin x}{1+\sin x}\right) + c$$

$$c) i) \frac{x^2}{16} + \frac{y^2}{9} = 1 \rightarrow a = 4, b = 3$$

$$\therefore \frac{2x}{16} + \frac{2y}{9} \cdot \frac{dy}{dx} = 0 \rightarrow \frac{dy}{dx} = -\frac{9x}{16y}$$

$$\therefore \text{ at } P(4\cos\theta, 3\sin\theta), \frac{dy}{dx} = -\frac{36\cos\theta}{48\sin\theta} = -\frac{3\cos\theta}{4\sin\theta}$$

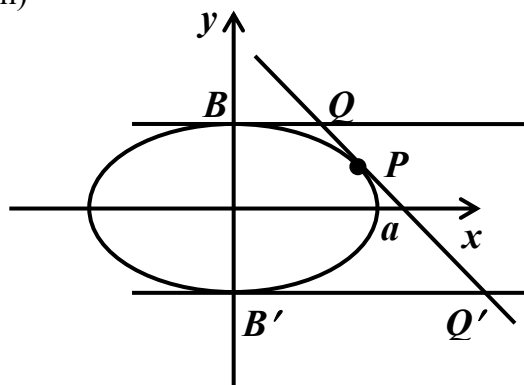
$$\therefore \text{ eq}^n \text{ of } T \text{ at } P: y - 3\sin\theta = -\frac{3\cos\theta}{4\sin\theta}(x - 4\cos\theta)$$

$$4y\sin\theta - 12\sin^2\theta = -3x\cos\theta + 12\cos^2\theta$$

$$3x\cos\theta + 4y\sin\theta = 12(\sin^2\theta + \cos^2\theta)$$

$$\therefore \frac{x\cos\theta}{4} + \frac{y\sin\theta}{3} = 1 \quad [\sin^2\theta + \cos^2\theta = 1]$$

13c)ii)



$$\text{At } Q: y = 3 \rightarrow x = \frac{4(1 - \sin\theta)}{\cos\theta}$$

$$\text{At } Q': y = -3 \rightarrow x = \frac{4(1 + \sin\theta)}{\cos\theta}$$

$$\begin{aligned} \therefore BQ \times B'Q' &= \frac{4(1 - \sin\theta)}{\cos\theta} \times \frac{4(1 + \sin\theta)}{\cos\theta} \\ &= \frac{16(1 - \sin^2\theta)}{\cos^2\theta} = 16 \end{aligned}$$

Question 14.

$$a) i) |x| + 1 = \frac{6}{|x|}$$

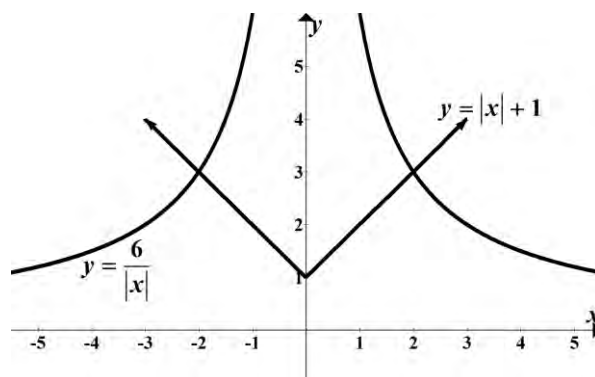
$$\therefore |x|^2 + |x| - 6 = 0$$

$$\therefore (|x| + 3)(|x| - 2) = 0$$

$$\therefore |x| = -3 \rightarrow \text{no solution}$$

$$\text{or } |x| = 2 \rightarrow x = \pm 2$$

ii)



iii) $-2 < x < 0$ and $0 < x < 2$

b) i) $(a-b)^2 \geq 0$
 $\therefore a^2 - 2ab + b^2 \geq 0$
 $\therefore a^2 + b^2 \geq 2ab$

ii) using (i) let $a = \sqrt{p}$ and $b = \sqrt{q}$
 $\therefore \sqrt{p}^2 + \sqrt{q}^2 \geq 2\sqrt{p}\sqrt{q}$
 $\therefore p + q \geq 2\sqrt{p}\sqrt{q}$ -----[1]

similarly let $a = \sqrt{p}$ and $b = \sqrt{2}$
 $\therefore p + 2 \geq 2\sqrt{p}\sqrt{2}$ -----[2]

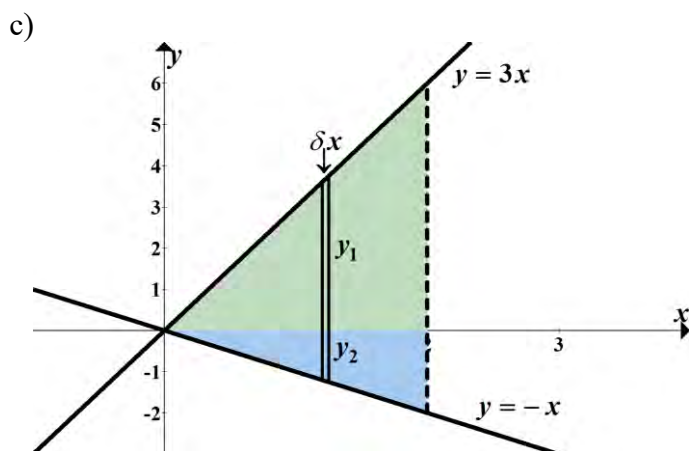
also let $a = \sqrt{q}$ and $b = \sqrt{2}$
 $\therefore q + 2 \geq 2\sqrt{q}\sqrt{2}$ -----[3]

hence [1] $\times$ [2] $\times$ [3] gives:

$$(p+2)(q+2)(p+q) \geq 2\sqrt{p}\sqrt{q} \times 2\sqrt{p}\sqrt{2} \times 2\sqrt{q}\sqrt{2}$$

$$\therefore (p+2)(q+2)(p+q) \geq 16(\sqrt{p})^2(\sqrt{q})^2$$

$$\therefore (p+2)(q+2)(p+q) \geq 16pq$$



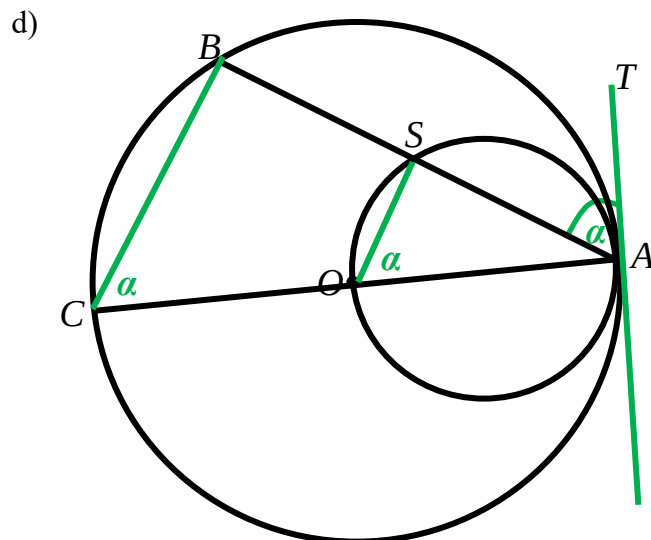
The area of a typical cross section is $\delta A = (y_1 + y_2)^2$

$$\therefore \delta A = (3x + x)^2 = 16x^2$$

$$\therefore \delta V = 16x^2 \delta x$$

$$\therefore V = 16 \int_0^2 x^2 dx$$

$$= 16 \left[\frac{x^3}{3} \right]_0^2 = \frac{128}{3} u^3$$



Construct tangent TA at A . Construct diameter AOC .
Join OS and BC .

Let $\angle TAS = \alpha$.

$$\therefore \angle SOA = \alpha \quad [\angle \text{ in alternate segment } (\Delta AOS)]$$

$$\therefore \angle BCA = \alpha \quad [\angle \text{ in alternate segment } (\Delta BCA)]$$

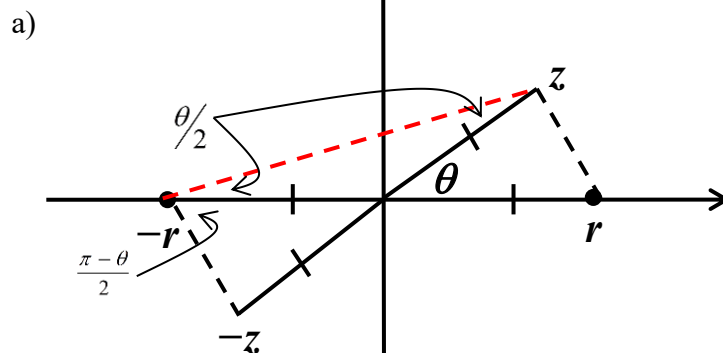
$\therefore \angle SOA = \angle BCA$ and since they are equal corresponding angles then BC is parallel to OS .

Also $OA = OC$ (equal radii)
and since a family of parallel lines divides transversals in the same ratio, then $BS = SA$.

Hence chord AB is bisected at S .

[Note: Congruent triangles can also be used.]

Question 15.



$$\arg(r^2 - z^2) = \arg(r - z) + \arg(r + z)$$

$$\text{Now } \arg(r + z) = \frac{\theta}{2} \quad (\text{by isosceles } \Delta)$$

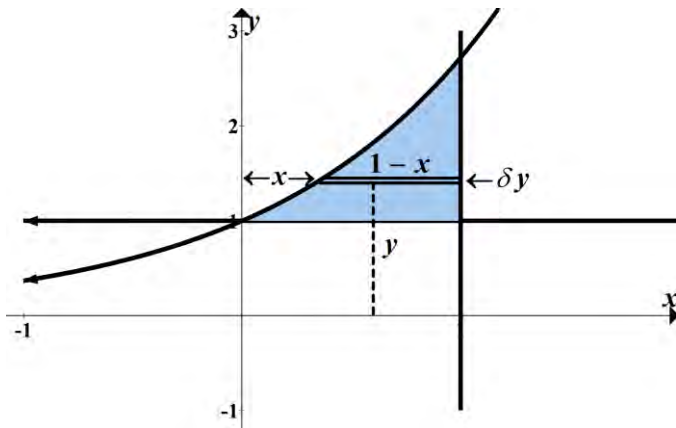
$$\text{also } \arg(r - z) = \frac{\theta}{2} - \frac{\pi}{2} \quad (\text{by isosceles } \Delta)$$

$$\therefore \arg(r^2 - z^2) = \frac{\theta}{2} - \frac{\pi}{2} + \frac{\theta}{2} = \theta - \frac{\pi}{2}$$

15b) i)

$$\begin{aligned}\frac{d}{dx}\left(\frac{x^2}{2}\ln x - \frac{x^2}{4}\right) &= \frac{x^2}{2} \times \frac{1}{x} + \frac{2x}{2} \times \ln x - \frac{2x}{4} \\ &= \frac{x}{2} + x \times \ln x - \frac{x}{2} = x \ln x\end{aligned}$$

ii)



Taking the middle of the shell at a distance 'y' units from the x-axis gives the following two radii:

$$R = y + \frac{\delta y}{2} \text{ and } r = y - \frac{\delta y}{2}$$

$$\therefore \text{Volume } \delta V \text{ of typical cylinder} = \pi(R+r)(R-r)h$$

$$= \pi\left(y + \frac{\delta y}{2} + y - \frac{\delta y}{2}\right)\left(y + \frac{\delta y}{2} - y + \frac{\delta y}{2}\right)h$$

$$= 2\pi y h \delta y \text{ where } h = 1 - x$$

$$\text{Now } y = e^x \rightarrow x = \ln y \text{ [when } x=1, y=e\text{]}$$

$$\therefore V = 2\pi \int_1^e y(1 - \ln y) dy$$

$$\therefore V = 2\pi \int_1^e y dy - 2\pi \int_1^e y \ln y dy$$

$$= 2\pi \left[\frac{y^2}{2} \right]_1^e - 2\pi \left[\frac{y^2}{2} \ln y - \frac{y^2}{4} \right]_1^e$$

$$= 2\pi \left(\frac{e^2}{2} - \frac{1}{2} \right) - 2\pi \left[\left(\frac{e^2}{2} \ln e - \frac{e^2}{4} \right) - \left(\frac{1}{2} \ln 1 - \frac{1}{4} \right) \right]$$

$$= \pi(e^2 - 1) - \pi \left(e^2 - \frac{e^2}{2} + \frac{1}{2} \right)$$

$$= \frac{\pi}{2}(e^2 - 3)$$

$$\begin{aligned}\text{c) i) } f(x) &= e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) \\ \therefore f'(x) &= e^{-x} \left(1 + \frac{2x}{2!} + \frac{3x^2}{3!} + \dots + \frac{nx^{n-1}}{n!} \right) \\ &\quad - e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right)\end{aligned}$$

$$\therefore f'(x) = e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^{n-1}}{(n-1)!} \right)$$

$$- e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right)$$

$$= -e^{-x} \left(\frac{x^n}{n!} \right) < 0 \text{ for all } x \geq 0.$$

$$\text{since } e^{-x} > 0 \text{ and } \frac{x^n}{n!} > 0.$$

hence the curve is decreasing for all $x \geq 0$

as $f'(x) < 0$.

ii) When $x=0$ $f(0) = 1$

and since the curve is always decreasing $f(x) \leq 1$.

$$\therefore e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) \leq 1$$

$$\therefore 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \leq \frac{1}{e^{-x}}$$

$$\text{hence } e^x \geq 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}.$$

Question 16.

$$\text{a) } \int \frac{e^{2x}}{e^x - 1} dx = \int \frac{e^x \cdot e^x}{e^x - 1} dx$$

$$\text{Let } u = e^x - 1 \rightarrow du = e^x dx$$

$$\therefore I = \int \frac{u+1}{u} du = \int 1 + \frac{1}{u} du$$

$$= u + \ln u$$

$$= e^x - 1 + \ln(e^x - 1) + C_1$$

$$= e^x + \ln(e^x - 1) + C_2$$

Question 16 continued:

b) i) $xy = c^2 \rightarrow y = c^2 x^{-1}$

$$\therefore y' = \frac{c^2}{x^2} \rightarrow m = -\frac{1}{p^2} \text{ when } x = cp$$

Equation of tangent at P :

$$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$$

$$p^2 y - cp = -x + cp$$

$$\therefore x + p^2 y = 2cp$$

ii) since OT is perpendicular tangent at P it has p^2 as its gradient.

$\therefore$ equation of OT is $y = p^2 x$, and since $T(x_1, y_1)$ lies on OT then $y_1 = p^2 x_1$.

iii) Since T satisfies the equation of the tangent:

$$x_1 + p^2 y_1 = 2cp$$

$$\text{and from (ii) } p^2 = \frac{y_1}{x_1}$$

$$\therefore x_1 + \frac{y_1}{x_1} \cdot y_1 = 2c \sqrt{\frac{y_1}{x_1}}$$

$$\therefore x_1^2 + y_1^2 = 2c x_1 \sqrt{\frac{y_1}{x_1}}$$

$$\therefore x_1^2 + y_1^2 = 2c \sqrt{x_1 y_1}$$

$$\therefore (x_1^2 + y_1^2)^2 = 4c^2 x_1 y_1$$

hence the locus of T is given by

$$(x^2 + y^2)^2 = 4c^2 xy.$$

c) i) $I_n = \int_0^1 x^{2n+1} e^{-x^2} dx = \int_0^1 x^{2n} \cdot x e^{-x^2} dx$

$$u = x^{2n}, \quad v' = x e^{-x^2}$$

$$u' = 2n x^{2n-1}, \quad v = -\frac{1}{2} e^{-x^2}$$

$$\therefore I_n = \left[-\frac{1}{2} x^{2n} e^{-x^2} \right]_0^1 + n \int_0^1 x^{2n-1} e^{-x^2} dx$$

$$= -\frac{1}{2} e^{-1} - 0 + n I_{n-1}$$

$$\therefore I_n = -\frac{1}{2e} + n I_{n-1} \text{ -----} \boxed{*}$$

ii) $I_0 = \int_0^1 x e^{-x^2} dx$

$$= \left[-\frac{1}{2} e^{-x^2} \right]_0^1 = -\frac{1}{2} (e^{-1} - e^0)$$

$$= -\frac{1}{2} \left(\frac{1}{e} - 1 \right) = \frac{1}{2} - \frac{1}{2e}$$

also $I_1 = -\frac{1}{2e} + I_0$

$$= -\frac{1}{2e} + \frac{1}{2} - \frac{1}{2e} = \frac{1}{2} - \frac{1}{e}$$

iii) When $n=1$: $LHS = 1 + \frac{1}{1!} = 2$

$$RHS = e - \frac{2e I_1}{1!}$$

$$= e - 2e \left(\frac{1}{2} - \frac{1}{e} \right)$$

$$= e - e + 2 = 2 = LHS$$

hence true for $n=1$

Assume true for $n=k$:

$$1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} = e - \frac{2e I_k}{k!}$$

Prove true for $n=k+1$:

i.e. $S_k + T_{k+1} = S_{k+1}$ (where $S_{k+1} = e - \frac{2e I_{k+1}}{(k+1)!}$)

$$LHS = e - \frac{2e I_k}{k!} + \frac{1}{(k+1)!}$$

$$= e - \frac{2e(k+1)I_k}{(k+1)!} + \frac{1}{(k+1)!}$$

$$= e + \frac{1 - 2e(k+1)I_k}{(k+1)!}$$

$$= e + \frac{1 - 2e \left(\frac{1}{2e} + I_{k+1} \right)}{(k+1)!}$$

$$= e + \frac{1 - 1 - 2eI_{k+1}}{(k+1)!}$$

$$= e - \frac{2eI_{k+1}}{(k+1)!} = S_{k+1}$$

Hence by induction is true for all $n \geq 1$.

NOTE: $I_{k+1} = -\frac{1}{2e} + (k+1)I_k$ using $\boxed{*}$

$$\therefore (k+1)I_k = \frac{1}{2e} + I_{k+1}$$

iv) $0 \leq I_n \leq 1$, hence as $n \rightarrow \infty$, $\frac{I_n}{n!} \rightarrow 0$

$$\therefore 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} = e - 2e \times 0 = e$$

$$\therefore \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots = e - 1$$